Habitat Characterization using Remote Image Sensory

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What it does

1. **Upload a drone orthomosaic**
   One big stitched drone image of any site.

2. **Our model finds the tree crowns**
   It outlines every crown it can detect in the image.

3. **Similar crowns are grouped**
   User get sample pictures of each group to review at on frontend, rest can be viewed and downloaded from shareable URL.

4. **You name each group once**
   Every tree in that group get labelled the species user inputted.

5. **Download your results**
   A species map, csv table summary and a stack item.

The Pipeline :

Orthomosaic goes in. A species map comes out. Eight steps, one of them which is labelling requires user inputs.

Stage 1 · Find the crowns (automatic)

1 · Orthomosaic

GeoTIFF from a drone survey, ideally finer than 8 cm/px.

2 · Rescale and tile

Brought to GSD best suited for the model, then cut into overlapping tiles.

3 · Detectree2

Mask R-CNN outlines the crowns in each tile, with a confidence score for each crown.

4 · Crown layer

Tiles stitched; weak, tiny, huge and duplicate crowns are removed.

Stage 2 · Group and name (automatic, then labelling)

5 · Crown crops

Each polygon is cut from the full-resolution image.

6 · DINOv2 + PCA

One embedding per crown, reduced to 50 dimensions.

7 · KMeans

Crowns grouped for every k; three scores suggest a k.

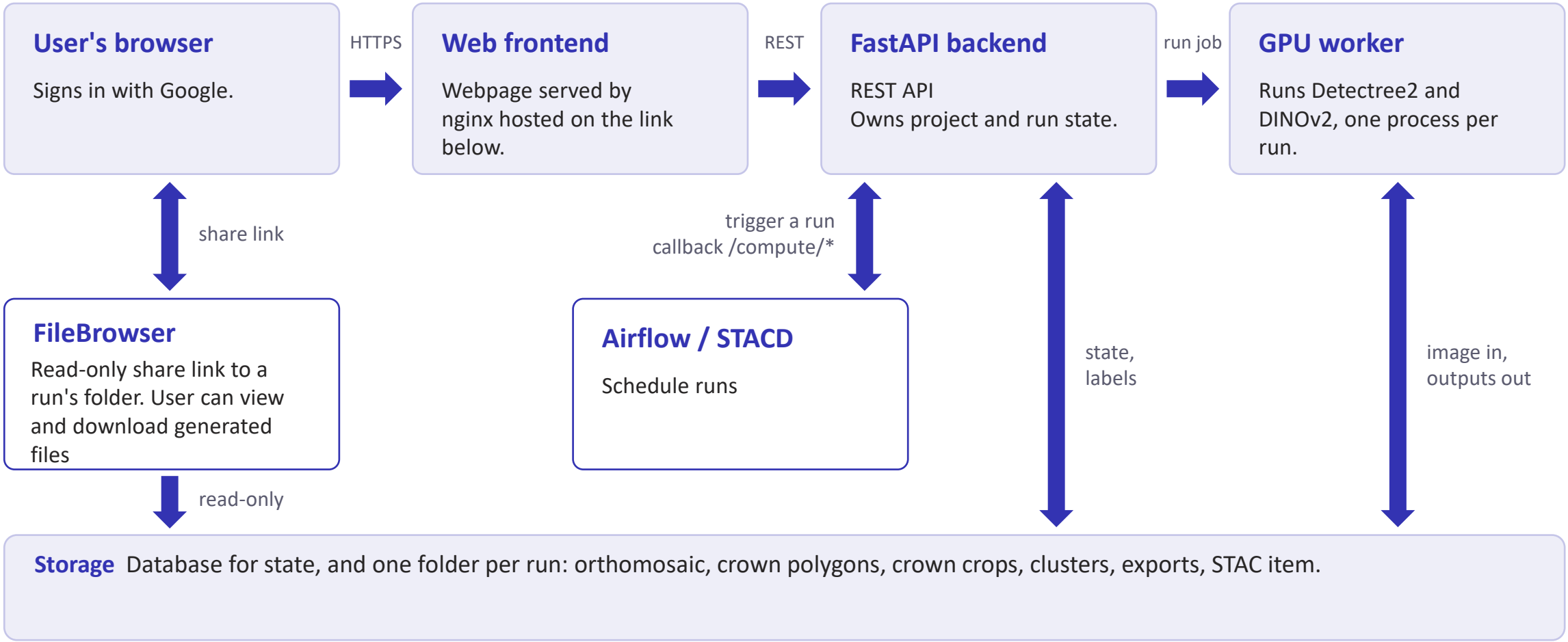
8 · User labels

The user after reviewing the crowns names each group and each tree crown in that group gets labelled.

Exports species map (KMZ) · per-crown CSV · crown polygons (GeoJSON) · STAC item

System architecture

The backend owns state, the worker computes, and the orchestrator only schedules.



Deployed at www.cse.iitd.ernet.in/act4dws5/drone

The Crown Detector

We run Detectree2, an open-source tree-crown detector.

What it is

- Mask R-CNN instance segmentation, built on Meta's detectron2.
- Each tile returns a crown outline and a confidence score
- Published weights are used as they are

Weights you can pick per project

urban_cambridge Urban trees, Cambridge UK. Default.

paracou Tropical canopy, Paracou UAV.

randresize Random-resize full model.

flexi Generalist: urban and closed canopy.

tropical_closed Dense tropical forest only.

The detector only finds crowns. User has to label the clusters according to their chosen value of k and final species map is generated as output.

Crown embedding: DINOv2

From crown to vector

- Each crown polygon is cut from the full-resolution orthomosaic.
- Then DINOv2 vision transformer returns one embedding per crown.
- Embeddings are L2-normalised, standardised, then reduced with PCA to 50 components(configurable).

Backbones you can pick per run

ViT-S/14 Fastest, smallest embeddings.

ViT-B/14 Balanced. Default.

ViT-L/14 Highest quality, slowest.

Clustering and choosing k

Look-alike crowns are grouped with KMeans, once for every k in the list.

KMeans for every k

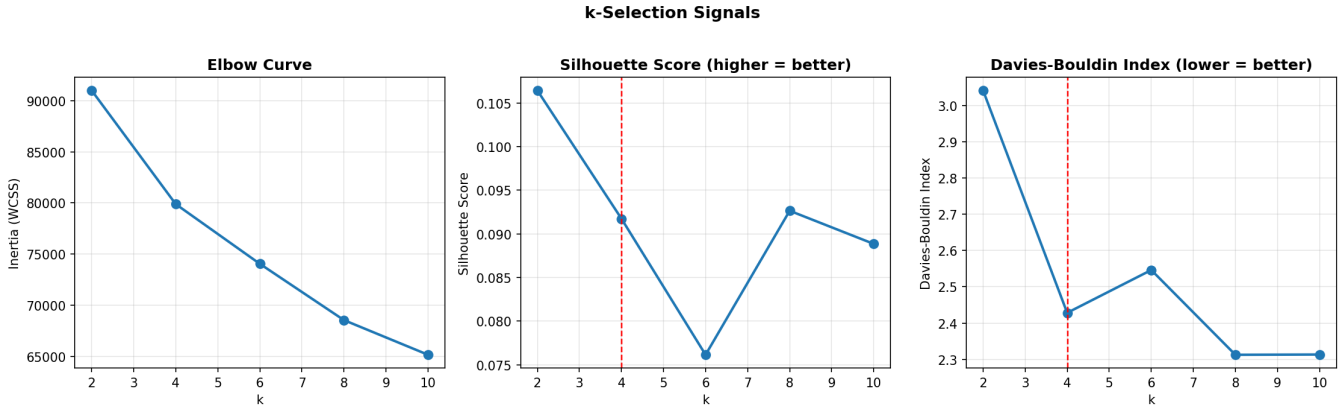
Default k list is 2, 4, 6, 8, 10.

Three scores

The elbow, silhouette and Davies-Bouldin are computed for every k.

Suggested k

The best k is suggested; the user can choose any k of their choice after reviewing.



The k-selection plot from a run. The red dashed line marks the suggested k.

For review, the app also shows the crowns nearest each group's centre and a t-SNE map of all crowns.

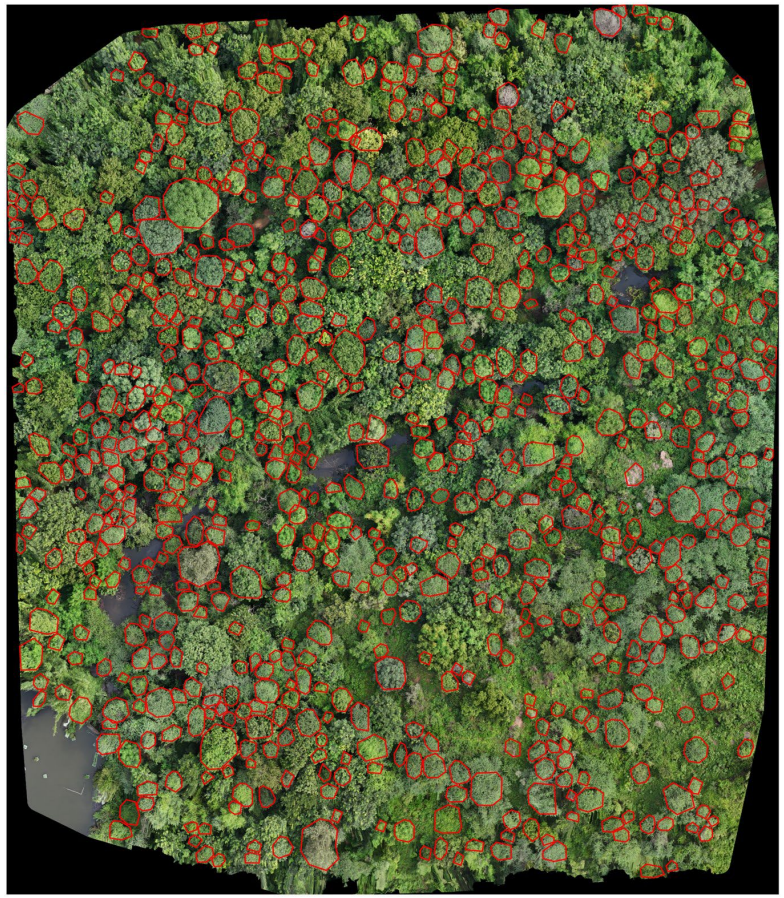
Naming species

Clustering only groups crowns. User supplies the species names after reviewing the cropped crowns.

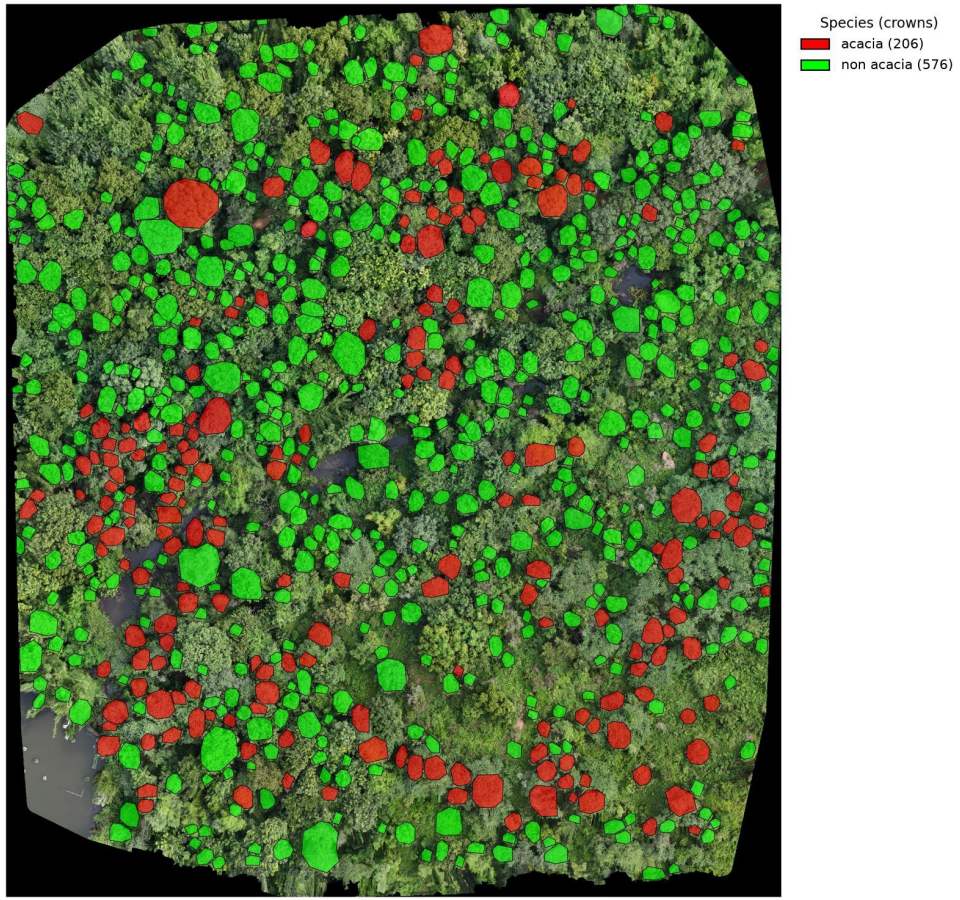
User names each group

- The review page shows sample crowns for every group at the chosen k.
- The user types one species per group; every crown in the group inherits it.
- Two groups can be given the same species.
- Labels are stored per run, so an older run can still be finished later.

Result



1 · Detected crowns



2 · Species map after labelling

782

tree crowns detected

206

labelled acacia (26%)

576

labelled non-acacia (74%)

One orthomosaic
2 cm/px, k = 2

Detectree2 (urban_cambridge) and DINOv2 ViT-B/14. Species come from user-named clusters and are not yet checked against ground truth.

What you get as Final Output

Species map (KMZ)

Opens in Google Earth or any GIS tool.

CSV tables

Every crown with its species, ready for Excel.

STAC record

So the dataset can be catalogued and found later.

FileBrowser folder

A shareable link to every file of the run.

Every run is kept

Compare settings and finish any run later.

